Sheet 1

DATE: __/__/____ **Assignment 1** DAY: __/__/____

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Course: Probability & Statistics
Program: BS CS
Session: F25 - Morning
Instructor: Ms. Iqra Aziz
Date of issue: 10/05/2025
Due Date: 10/06/2025

Data:

Class Interval	Frequency
11-20	13
21-30	14
31-40	23
41-50	33
51-60	07

Q1. Find

a) Mean
b) Median
c) Mode
d) Standard Deviation
e) Variance

Solution:-

a) **Mean ($\bar{x}$)**

Formula = $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

DATE: __/__/____ DAY: __/__/____

Class Interval	f	x	fx
11 - 20	13	$\frac{11+20}{2} = 15.5$	201.5
21 - 30	14	$(21+30)/2 = 25.5$	357
31 - 40	23	$(31+40)/2 = 35.5$	816.5
41 - 50	33	$(41+50)/2 = 45.5$	1501.5
51 - 60	07	$(51+60)/2 = 55.5$	388.5

$\Sigma f = 90$ $\Sigma fx = 3265$

$$\boxed{\bar{x} = \frac{3265}{90} = 36.28}$$

b. Median ($\tilde{x}$)

Formula

$$\tilde{x} = L + \frac{h}{f}\left[\frac{n}{2} - C\right]$$

Solution:-

CI	f	C.B	Cf
11-20	13	10.5-20.5	13
21-30	14	20.5-30.5	27
31-40	(f) 23	(L) 30.5-40.5	50 (c)
41-50	33	40.5-50.5	83
51-60	07	50.5-60.5	90

$\Sigma f = 90$

let put known values in formula

$\therefore L = 30.5$

$\therefore h = 10$ (interval size)

$\therefore f = \cancel{\Sigma f = 90}\ 23$

$\therefore n = \Sigma f = 90$

$\therefore c = 50$

DATE: __/__/__ DAY: __/__/__

So

$$\tilde{x} = 30.5 + \frac{10}{23}\left[\frac{90}{2} - 50\right]$$

$$= 30.5 + \frac{5}{23}(90-50)$$

$$= 30.5 + \frac{5}{23}40$$

$$= 30.5 + \frac{200}{23}$$

$$= \frac{701.5 + 200}{23}$$

$$= \frac{901.5}{23}$$

$$= 39.196$$

$$= 30.5 + \frac{10}{23}(-50)$$

$$= 30.5 - \frac{50}{23}$$

$$= \frac{701.5 - 50}{23}$$

$$= \frac{651.5}{23}$$

$$= 28.33$$

$$\boxed{\tilde{x} = 28.33}$$

C) Mode ($\hat{x}$)

formula $\hat{x} = L + \frac{(f_m - f_1)}{(2f_m - f_1 - f_2)} h$

ution:-

CI	f	CB
11-20	13	10.5-20.5
21-30	f_1 14	20.5-30.5
31-40	f_m 23	L 30.5-40.5
41-50	f_2 33	40.5-50.5
51-60	07	50.5-60.5

Known data

$L = 30.5$

$f_m = 23$

$f_1 = 14$

$f_2 = 33$

$h = 10$ (interval size)

DATE: __/__/____ DAY: __/__/____

Now putting all known data into formula, we get

$$\hat{x} = \frac{30.5 + (23-14)}{2(23) - 14 - 23}(10)$$

$$= \frac{30.5 + 9}{46 - 37}(10)$$

$$= \frac{39.5}{9}(10)$$

$$= \frac{395}{9} = 43.89$$

$$\boxed{\hat{x} = 43.89}$$

d) Standard Deviation

formula

$$\sigma = \sqrt{\frac{\Sigma f(x-\bar{x})^2}{n \Sigma f}}$$

Required to find

x = mean of each interval

$\bar{x}$ = mean of x

n = Σf

CI	f	x	fx	$x-\bar{x}$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
10-20	13	15.5	201.5	-20.78	431.81	5613.53
21-30	14	25.5	357	-10.78	116.2	1626.8
31-40	23	35.5	816.5	-0.78	0.61	14.03
41-50	33	45.5	1501.5	9.22	85.01	2805.33
51-60	07	55.5	388.5	19.22	369.41	2585.87

$\Sigma f = 90$ $\Sigma fx = 3265$ $\cancel{\Sigma = 1003.04}$ $\Sigma = 12645.5$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{3265}{90} = 36.28$$

$$S = \sqrt{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}}$$

$$= \sqrt{\frac{\cancel{1003.04}\ 12645.56}{90}}$$

$$= \sqrt{\cancel{11.145}\ 140.51}$$

$$\boxed{S = \cancel{3.34}\ 11.85}$$

e) Variance

formula

$$S^2 = \frac{\Sigma f(x-\bar{x})^2}{\Sigma f}$$

Putting all known values in formula, we get

DATE: __/__/____ DAY: __/__/____

$$\boxed{S^2 = 140.51.}$$

Q.2. Key Pros & Key Cons

Mean

Pros	Cons
→ Easy to understand & calculate	→ Affected by extreme observation
→ No need to arrange data	→ Graphical representation is not possible
→ least affected by sampling fluctuation, hence more stable	→ Can be used only for quantitative data

Median

Pros	Cons
→ Not affected by extreme observation	→ Affected more by sampling fluctuations
→ for both qualitative & quantitative data	→ Equal importance to all data
→ Graphical presentation is possible	→ Need to arranging data i.e. asc or desc.

Mode

Pros	Cons
→ Not affected by extreme observation	→ Not rigidly define

DATE: __/__/____ DAY: __/__/____

→ Both for quantitative & qualitative data → Graphical representation is possible.	→ Can be used for future mathematical calculations

Standart Deviation

pros	Cons
→ A precise measure of data variability around mean. → Based on all values in dataset.	→ Sensitive to outliers, similer to the mean → more complex to calculate than measures of central tendency.

Variance

Pros	Cons
→ Considers the full range of risk & variability by treating all deviations from the mean equally. → Useful for understanding the spread or dispersion of data.	→ provides a measure of spread, not a typical or central value. → Doesn't indicate the direction of deviations, only their magnitude.

-(THE END)-